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***Abstract*—Drone-view object detection is fundamental to applications ranging from aerial surveillance to precision agriculture. However, the dense distribution of small objects in drone imagery makes traditional bounding-box annotation prohibitively labor-intensive. Although Point-Supervised Object Detection (PSOD) offers a cost-effective alternative, existing methods still face major challenges: teacher-student frameworks often require heavy computation and conservative supervision, while early pseudo-label-based pipelines typically depend on dataset-specific hyperparameters and heuristic priors. In addition, the conventional two-stage pipeline—separating pseudo-label generation from detector training—introduces extra complexity and computational overhead. To address these issues, we propose an efficient end-to-end PSOD framework with a variation-based map. Our method adaptively estimates object extent through statistical variation modeling, reducing manual hyperparameter tuning while improving robustness in dense scenes.**

***Index Terms*—point-supervised object detection, drone-view detection, oriented object detection, variational inference**

I. Introduction

Drone-view object detection, which aims to identify targets in UAV-captured imagery, plays a vital role in applications such as aerial surveillance, traffic monitoring, and precision agriculture.

However, annotating drone-view images with bounding boxes is highly inefficient and labor-intensive because such images often contain many densely packed small targets, making annotation substantially more difficult than in ground-view images. Therefore, low-cost and efficient annotation paradigms are crucial. Recently, Point-Supervised Object Detection (PSOD) has attracted significant attention because point annotations are much cheaper and faster to collect than bounding-box annotations.

Most early PSOD methods can be viewed as teacher-student frameworks, where a teacher branch iteratively provides supervisory signals to a student detector. In dense scenes, these frameworks often suffer from conservative sample assignment and unstable pseudo-supervision updates.

Following the analysis in SSP [1], pseudo-label-based frameworks provide two key advantages over teacher-student frameworks: (1) higher training efficiency, because they avoid the duplicated optimization cost of teacher-student pipelines; and (2) higher label utilization, because they can exploit point-layout priors to mine richer positive/negative supervision beyond fixed-radius assignment.

PointOBB-v2 [2] is a representative pseudo-label-based pipeline that explicitly decouples pseudo-box generation from detector optimization. As summarized by SSP [1], it typically uses a $0.5\times$ stage for pseudo-label generation and a $1\times$ stage for detector training, whereas teacher-student pipelines are often closer to $2\times1\times$ optimization. Starting from single-point

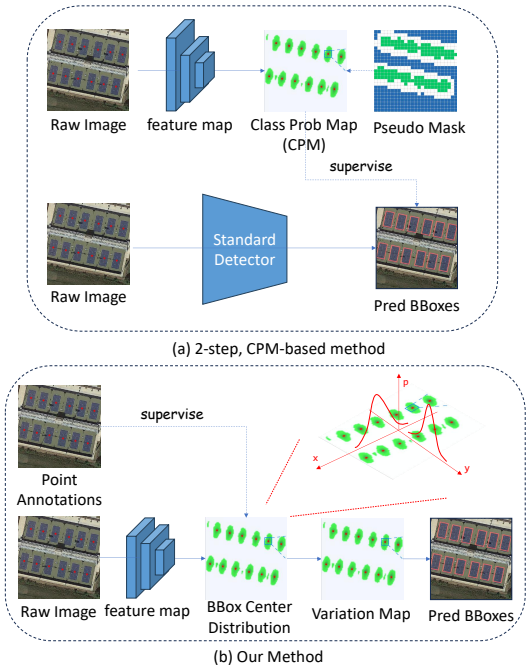

Fig. 1. Comparison between CPM-based pseudo-box generation and the proposed variation-based map. CPM may underestimate object extent when low confidence suppresses activation, while the variation map better separates uncertain boundaries from reliable object interiors.

annotations, PointOBB-v2 first trains a point-supervised classifier with non-uniform positive/negative sampling to produce a class probability map (CPM), where high-response regions approximately indicate object support. It then applies Principal Component Analysis (PCA) on activated CPM regions to infer oriented geometry (center, major/minor axes, and angle), and introduces a separation mechanism to alleviate ambiguity when adjacent instances overlap heavily on the map. This design substantially improves speed-accuracy trade-offs, but its quality still depends on handcrafted hyperparameters (e.g., negative/positive label radius), which can be sensitive to density patterns across datasets.

However, CPM-style supervision still has an intrinsic ambiguity in point-supervised settings: low confidence usually leads to low activation, and low activation may be misinterpreted as physically small object extent rather than epistemic uncertainty. As a result, boundaries of large-but-hard objects can be over-shrunk, especially under occlusion or clutter, and decision quality is highly affected by handcrafted hyperparameters for thresholding and region partition.

In this paper, we propose a variation-based map that decouples uncertainty from spatial extent. It avoids dataset-

specific handcrafted hyperparameters for size inference and provides clearer boundary cues by assigning high variation to ambiguous boundary/background regions while keeping object interiors relatively stable. As a result, the proposed map is more robust to confidence fluctuations and better aligned with true object geometry in dense drone-view scenes.

Another limitation of previous PSOD methods is the two-stage training paradigm. In the first stage, a pseudo-label generator predicts pseudo-boxes from point annotations. In the second stage, a standard box detector is trained on the pseudo-labeled dataset. This decoupled process increases training cost, introduces extra optimization stages, and may propagate pseudo-label errors from the first stage to the final detector.

Recent progress has shown that end-to-end point-supervised frameworks can reduce this mismatch by jointly optimizing pseudo-box inference and detection objectives in a unified pipeline, leading to simpler training and better supervision consistency. However, existing end-to-end variants still need stronger uncertainty modeling in dense drone scenes, where point ambiguity, occlusion, and boundary overlap remain challenging.

To address these limitations, we propose an effective end-to-end point-supervised framework with a variation-based map that jointly optimizes uncertainty-aware extent estimation and detector learning in a single training process.

Our key contributions are summarized as follows.

- We introduce an uncertainty-aware variation map derived from the posterior uncertainty of latent variables. Through Gaussian parameterization and reparameterization-based optimization, the method adaptively models object extent without dataset-specific handcrafted spatial hyperparameters.
- We propose an end-to-end point-supervised object detection framework that unifies latent extent inference and detector learning in a single training pipeline, removing the conventional two-stage pseudo-label generation procedure.
- Extensive experiments on the challenging drone-view dataset DOTA-v1.0 show that the proposed method consistently outperforms representative teacher-student and pseudo-label-based PSOD baselines while preserving high training and inference efficiency.

II. Related Works

A. Teacher-student Frameworks

Early PSOD studies are often organized as teacher-student frameworks, where supervisory signals are progressively refined and passed to the final detector. Representative systems such as P2BNet [3], LIFOD [4], and PointOBB [5] illustrate this paradigm. Related studies such as One-Point WSOD [6] and PointSR [7] also highlight the importance of robust point-driven supervision. However, these frameworks are often bottlenecked by conservative sample assignment and additional optimization overhead.

B. Pseudo-label-based Frameworks

To bypass teacher-student optimization overhead, pseudo-label-based frameworks generate pseudo-boxes directly from point-supervised maps. SSP [1] emphasizes two key advantages of this paradigm over teacher-student frameworks: higher training efficiency and higher label utilization. PointOBB-v2 [2] is a representative method: it first predicts dense class probability maps from point supervision, then extracts oriented pseudo-boxes from high-response regions (e.g., via geometric fitting/PCA), and finally trains a detector on these pseudo-labels. This design is simpler and faster, but it still relies on handcrafted post-processing choices (such as response thresholding and region separation) that can be sensitive across datasets. SSP [1] further follows this paradigm and introduces partition-based strategies for more stable sample assignment and pseudo-label extraction.

C. End-to-End

Recent progress has shifted toward end-to-end (E2E) learning, which unifies pseudo-label generation and detector training. Methods such as Point2RBox [8] and PointOBB-v3 [9] further demonstrate the potential of this direction; additional point-supervised tiny-object studies (e.g., PointTeacher [10]) are also aligned with this trend. This design reduces error propagation inherent to two-stage pipelines and streamlines deployment while improving efficiency.

III. Proposed Method

We formulate point-supervised object detection as a variational inference problem. Given an input image and point annotations, we infer a latent proposal distribution and optimize the detector by maximizing expected detection likelihood under this distribution.

A. Problem Definition

Let $\mathbf{f}$ denote the backbone feature map, $\mathbf{p}$ denote point annotations, and $\mathbf{y}$ denote detection targets (class and oriented-box parameters). We introduce latent variables $\mathbf{z}$ to represent uncertainty-aware proposal states (center offset, scale, and orientation). The prior is not assumed to be a simple isotropic Gaussian; instead, we construct a ground-truth-conditioned discrete prior to encode instance-aware geometric statistics.

The conditional likelihood is

$$\log p_\theta(\mathbf{y} \mid \mathbf{f}, \mathbf{p}) = \log \int p_\theta(\mathbf{y} \mid \mathbf{z}, \mathbf{f})\, p_\psi(\mathbf{z} \mid \mathbf{f}, \mathbf{p})\, d\mathbf{z}, \quad (1)$$

which is intractable. We therefore optimize an ELBO with an approximate posterior $q_\phi(\mathbf{z} \mid \mathbf{f}, \mathbf{p})$:

$$\mathcal{L}_{\text{ELBO}} = \mathbb{E}_{q_\phi}\big[\log p_\theta(\mathbf{y} \mid \mathbf{z}, \mathbf{f})\big] - D_{\text{KL}}\big(q_\phi(\mathbf{z} \mid \mathbf{f}, \mathbf{p}) \| p_\psi(\mathbf{z} \mid \mathbf{f}, \mathbf{p})\big). \quad (2)$$

B. Variational Detection

Posterior parameterization. We use a Gaussian posterior to model latent proposal uncertainty:

$$q_\phi(\mathbf{z} \mid \mathbf{f}, \mathbf{p}) = \mathcal{N}\big(\boldsymbol{\mu}_\phi(\mathbf{f}, \mathbf{p}),\, \boldsymbol{\sigma}^2_\phi(\mathbf{f}, \mathbf{p})\big). \quad (3)$$

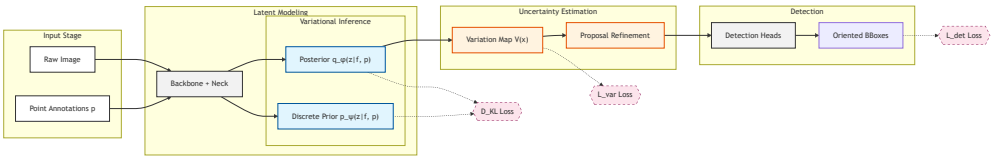

Fig. 2. Overall pipeline of the proposed framework. Given an input image with point annotations, the model learns a variational posterior for proposal states, aligns it with a ground-truth-conditioned discrete prior, and builds an uncertainty-aware variation map for proposal refinement and final oriented detection.

Ground-truth-conditioned prior. We replace the conventional Gaussian prior with a ground-truth-conditioned discrete prior. Concretely, we discretize box geometry into bins and construct an instance-aware prior distribution from the target geometry:

$$p_\psi(\mathbf{z} \mid \mathbf{f}, \mathbf{p}) = \mathrm{Cat}\big(\boldsymbol{\pi}_{\mathrm{gt}}(\mathbf{p})\big), \tag{4}$$

where $\boldsymbol{\pi}_{\mathrm{gt}}$ denotes the discrete probability vector derived from ground-truth-conditioned statistics. Compared with Gaussian priors, this prior better captures asymmetric and multimodal geometry in crowded scenes, explicitly correlates latent proposals with object-specific extent, and reduces the mismatch caused by oversimplified unimodal assumptions.

Uncertainty-aware variation map. From posterior variance, we construct

$$\mathbf{V}(x) = \sigma^2(x \mid \mathbf{f}, \mathbf{p}), \tag{5}$$

where lower variance indicates reliable object interiors and higher variance indicates boundaries/background. Unlike CPM-style activation maps, which may confound low confidence with small object extent, $\mathbf{V}$ explicitly models uncertainty and geometry as disentangled signals. Therefore, low confidence does not directly shrink the predicted extent; instead, ambiguous regions are reflected as high variation near boundaries/background. This property improves boundary discrimination in crowded scenes and reduces sensitivity to hand-crafted post-processing hyperparameters (e.g., threshold/radius choices). We use $\mathbf{V}$ for proposal refinement and positive-sample reweighting.

C. Variational Optimization

Reparameterization and prior alignment. We keep the continuous posterior sampling for stable optimization:

$$\mathbf{z} = \boldsymbol{\mu}_\phi + \boldsymbol{\sigma}_\phi \odot \boldsymbol{\epsilon}, \quad \boldsymbol{\epsilon} \sim \mathcal{N}(0, \mathbf{I}). \tag{6}$$

Training objective. The final optimization objective is

$$\mathcal{L} = \mathcal{L}_{\mathrm{det}} + \lambda_{\mathrm{KL}} D_{\mathrm{KL}}\big(q_\phi(\mathbf{z} \mid \mathbf{f}, \mathbf{p}) \| p_\psi(\mathbf{z} \mid \mathbf{f}, \mathbf{p})\big) + \lambda_{\mathrm{var}} \mathcal{L}_{\mathrm{var}}, \tag{7}$$

where $\mathcal{L}_{\mathrm{det}}$ is the standard point-supervised detection loss and $\mathcal{L}_{\mathrm{var}}$ is the uncertainty-aware reweighting term induced by $\mathbf{V}$. The KL term regularizes the inferred posterior toward the ground-truth-conditioned discrete prior, which is more expressive than a fixed Gaussian prior for skewed or multi-peak target geometry. This objective encourages robust extent learning without dataset-specific handcrafted hyperparameters and improves geometric consistency under confidence fluctuations. At inference, we use the posterior mean for deterministic decoding.

D. Complexity Analysis

Compared with a standard detector, our method introduces lightweight distribution heads for posterior parameters and discrete-prior alignment. The additional complexity is linear in feature resolution and discrete bin count B:

$$\Delta\mathrm{FLOPs} = \mathcal{O}(HW \cdot C \cdot B), \tag{8}$$

which is small relative to backbone and neck computation. Therefore, the framework preserves practical training/inference efficiency while improving robustness in crowded drone-view scenes.

IV. Results

A. Experimental Settings

Datasets and metrics. We evaluate the proposed method on DOTA-v1.0 under point-level supervision. We report mAP, AP50, and AP75 as primary accuracy metrics, and additionally report efficiency indicators (Params, FLOPs, FPS, and training time).

Implementation protocol. For fair comparison, all methods use the same data split and training schedule whenever possible. Our method adopts the variational formulation in Sec. 3 with a configurable posterior parameterization, KL weight λ_{KL}, and latent dimension.

B. Benchmark Results

1) Main Comparison with State-of-the-Art: Table I compares our method with representative teacher-student, pseudo-label-based, and end-to-end PSOD methods, including PointOBB-v2 [2], Point2RBox [8], and PointOBB-v3 [9]. Numerical results will be reported in the final version.

2) Results under Occlusion Levels: Following the motivation of variational detection in crowded scenes, we further report performance under different occlusion levels.

qualitative_comparison_placeholder.pdf

Fig. 3. Qualitative comparison on DOTA-v1.0. From left to right: input image, baseline prediction, and our prediction. Our method shows clearer boundaries and better localization under dense layouts and occlusion.

TABLE I
MAIN COMPARISON ON DOTA-V1.0 UNDER POINT SUPERVISION.

Method	mAP	AP50	AP75
(Teacher-student baseline A)			
(Teacher-student baseline B)			
PointOBB-v2			
PointOBB-v3			
Ours (VAE-based)			

TABLE II
PERFORMANCE UNDER DIFFERENT OCCLUSION LEVELS (FILL WITH YOUR SPLIT PROTOCOL).

Method	Low Occ.	Mid Occ.	Heavy Occ.
PointOBB-v2			
Point2RBox			
PointOBB-v3			
Ours			

C. Ablation Study

1) Component Ablation: We first evaluate the contribution of each component in the proposed pipeline, including the variation map, boundary refinement, and discrete-prior regularization. Starting from an end-to-end baseline, we incrementally

occlusion_visualization_placeholder.pdf

Fig. 4. Visualization under different occlusion levels (low, medium, heavy). The proposed method maintains more stable extent estimation and orientation prediction in heavily occluded scenes.

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Fig. 5. Coordinate-wise effect of variational inference (VPD-style analysis). We compare different coordinate groups (only x, only y, and (x, y)) to show where uncertainty modeling contributes most. The gain is typically larger when both center coordinates are modeled variationally.

TABLE III
ABLATION EXPERIMENTS WITH VARIATIONAL INFERENCE ON DIFFERENT COORDINATES.

Variational coordinates	mAP	AP50	AP75
Deterministic baseline			
x only			
y only			
(x, y)			

TABLE IV
ABLATION ALIGNED WITH INTRODUCTION CLAIMS.

Setting	mAP	AP50	AP75
E2E baseline (CPM-based cue)			
+ Variation map			
+ Boundary refinement			
+ Discrete prior reg.			
+ Full model			

add each module to quantify its independent and cumulative impact.

2) Ablation Experiments with Variational Inference on Different Coordinates: Following the coordinate-wise analysis protocol used in prior variational detection studies, we investigate where variational inference contributes most to oriented bounding-box prediction. We progressively enable variational inference on different coordinate groups, including only x, only y, and both (x, y). This study evaluates whether uncertainty modeling primarily benefits center localization under dense and occluded conditions.

The coordinate-wise trend in Fig. 5 complements Table III. We observe that applying variational inference to full center coordinates (x, y) yields the most significant improvements in dense scenes, indicating that uncertainty-aware extent modeling primarily benefits spatial localization robustness.

We design ablations around three key questions: (1) whether the variation-based map alleviates the CPM ambiguity (low confidence vs. small extent), (2) whether boundary discrimination improves in dense/occluded scenes, and (3) whether the ground-truth-conditioned discrete prior reduces reliance on handcrafted spatial hyperparameters while keeping end-to-end optimization effective.

3) Different Reparameterization Strategies:

4) Efficiency and Sensitivity: To support the practicality claim in the Introduction, we report model complexity and speed and analyze sensitivity to key learning hyperparameters (λ_{KL}, latent dimension). Unlike CPM-style handcrafted spatial thresholds/radii, these optimization hyperparameters remain stable across datasets in our end-to-end setting.

5) Disentangling Effect in Dense Scenes: We visualize predicted oriented boxes together with the learned uncertainty map $\sigma^2(x \mid \mathbf{f}, \mathbf{p})$. Consistent with variational intuition, reliable object interiors tend to exhibit lower variance, while ambiguous boundaries and cluttered background show higher variance. This behavior helps separate adjacent dense instances and improves localization robustness.

6) Variational Effect on Localization Stability: Compared with deterministic point-supervised regression, variational modeling smooths optimization by marginalizing over latent proposal uncertainty. Empirically, this reduces unstable updates in crowded regions and improves AP75, indicating more accurate oriented box localization.

TABLE V
ABLATION ON POSTERIOR–PRIOR ALIGNMENT STRATEGY.

Strategy	mAP	AP50	AP75
No prior alignment			
Continuous posterior + Gaussian prior			
Continuous posterior + discrete prior			

efficiency_tradeoff_placeholder.pdf

Fig. 6. Efficiency–accuracy trade-off. Comparison of mAP versus inference speed (FPS) among representative point-supervised detectors. The proposed method aims to achieve a favorable Pareto balance.

7) Robustness to Occlusion and Noise: The variational detector is expected to be more robust when point annotations are noisy or targets are heavily occluded, because posterior inference can absorb annotation ambiguity through uncertainty modeling. Typical failure cases remain in extremely tiny targets and severe mutual occlusion, where all nearby proposals are highly ambiguous.

8) Practical Implications and Limitations: The proposed method remains lightweight and compatible with common detection pipelines, but it introduces additional hyperparameters (e.g., B, λ_{KL}). Future work includes adaptive prior construction, temporal consistency for drone videos, and uncertainty-guided active point annotation.

V. CONCLUSION

In this paper, we present an end-to-end point-supervised object detection framework with a variation-based map for dense drone-view scenes. The proposed method addresses key limitations of prior pseudo-label-based and teacher-student pipelines by disentangling uncertainty from object extent, reducing dependence on handcrafted spatial hyperparameters, and introducing a ground-truth-conditioned discrete prior for more expressive variational regularization. This design improves boundary discrimination and localization robustness under occlusion and crowding while preserving practical efficiency.

TABLE VI
EFFICIENCY COMPARISON.

Method	Params (M)	FLOPs (G)	FPS	Training Time (h)
CPM-based baseline				
End-to-end baseline				
Ours (VAE-based)				

TABLE VII
SENSITIVITY ANALYSIS OF KEY HYPERPARAMETERS.

Hyperparameter setting	mAP	AP50	AP75
$\lambda_{\mathrm{KL}} = 0.1$			
$\lambda_{\mathrm{KL}} = 0.5$			
$\lambda_{\mathrm{KL}} = 1.0$			
Latent dim $= 8$			
Latent dim $= 16$			
Latent dim $= 32$			

Extensive experiments on DOTA-v1.0 demonstrate consistent gains over representative point-supervised baselines, especially in high-precision localization and challenging dense scenarios. Ablation studies further verify that (1) variation-based modeling alleviates the low-confidence/low-activation ambiguity, (2) coordinate-wise variational inference brings complementary gains, and (3) discrete-prior alignment improves geometric consistency in end-to-end optimization.

In future work, we will explore adaptive prior construction, temporal uncertainty modeling for drone videos, and uncertainty-guided active point annotation to further improve annotation efficiency and detection robustness.

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REFERENCES

[1] Xinyaun Liu, Hang Xu, Yike Ma, Yucheng Zhang, and Feng Dai, "Semantic-decoupled spatial partition guided point-supervised oriented object detection," *arXiv preprint arXiv:2506.10601*, 2025.

[2] Botao Ren, Xue Yang, Yi Yu, Junwei Luo, and Zhidong Deng, "Pointobb-v2: Towards simpler, faster, and stronger single point supervised oriented object detection," *arXiv preprint arXiv:2410.08210*, 2024.

[3] Zhen Chen, Yuliang Wang, Gang Wang, Yu Qiao, and Huchuan Lu, "Point-to-box network for accurate object detection via single point supervision," in *European Conference on Computer Vision (ECCV)*, 2022.

[4] Zhe Luo, Dong Zhang, Wei Liu, and Jun Wang, "End-to-end point supervised object detection with low-level instance features," *Applied Soft Computing*, vol. 154, pp. 111350, 2024.

[5] Xue Yang, Yi Yu, Botao Ren, Junwei Luo, Zhidong Deng, and Junchi Yan, "Pointobb: Towards single point supervised oriented object detection," *arXiv preprint arXiv:2311.12816*, 2023.

[6] Cheng He, Yifan Li, Rui Zhao, and Fei Wang, "One point is all you need for weakly supervised object detection," *Pattern Recognition*, vol. 158, pp. 111045, 2025.

[7] Minghao Li, Yifan Zhang, Zekai Xu, Yong Liu, and Wenqi Luo, "Pointsr: Self-regularized point supervision for drone-view object detection," in *Proceedings of the IEEE/CVF Conference on Computer Vision and Pattern Recognition (CVPR)*, 2025.

[8] Yi Yu, Xue Yang, Qingyun Li, Feipeng Da, Jifeng Dai, Yu Qiao, and Junchi Yan, "Point2rbox: Combine knowledge from synthetic visual patterns for end-to-end oriented object detection with single point supervision," *arXiv preprint arXiv:2311.14758*, 2023.

[9] Peiyuan Zhang, Junwei Luo, Xue Yang, Yi Yu, Qingyun Li, Yue Zhou, Xiaosong Jia, Xudong Lu, Jingdong Chen, Xiang Li, Junchi Yan, and Yansheng Li, "Pointobb-v3: Expanding performance boundaries of single point-supervised oriented object detection," *arXiv preprint arXiv:2501.13898*, 2025.

[10] Han Yao, Liang Zhang, Jiaqi Peng, and Rui Wu, "Tiny object detection with single point supervision," *ISPRS Journal of Photogrammetry and Remote Sensing*, vol. 224, pp. 120–134, 2025.